A4

Q1

(i): $E\left(\min\{Y, X_1, \ldots, X_n\}^{-1/2}\right)$

Define $Z = \min\{Y, X_1, \ldots, X_n\}$.

Consider the minimum of the X_i 's alone first. For $X_1, \ldots, X_n$ (the probability density function is $f_{X_i}(x) = \theta e^{-\theta x}$ for $x > 0$), the minimum $M = \min\{X_1, \ldots, X_n\}$ is also exponential. The rate of this minimum is the sum of the individual rates, because the survival function is:

$$P(M > t) = P(X_1 > t, \ldots, X_n > t) = \prod_{i=1}^{n} P(X_i > t) = (e^{-\theta t})^n = e^{-n\theta t},$$

which is the survival function of an exponential distribution with rate $n\theta$. Thus, $M \sim \text{Exp}(n\theta)$.

Now, $Z = \min\{Y, M\}$, where $Y \sim \text{Exp}(\mu)$ and $M \sim \text{Exp}(n\theta)$, and they are independent because Y is independent of all X_i 's. For two independent exponentials with rates λ_1 and λ_2, the minimum has rate $\lambda_1 + \lambda_2$, since:

$$P(\min\{Y, M\} > t) = P(Y > t)P(M > t) = e^{-\mu t}e^{-n\theta t} = e^{-(\mu + n\theta)t}.$$

Thus, $Z \sim \text{Exp}(\mu + n\theta)$, with density $f_Z(z) = (\mu + n\theta)e^{-(\mu+n\theta)z}$ for $z > 0$.

For a random variable $Z \sim \text{Exp}(\lambda)$ with $\lambda = \mu + n\theta$:

$$E[Z^{-1/2}] = \int_0^\infty z^{-1/2}\lambda e^{-\lambda z}\, dz.$$

This integral resembles the form $\int_0^\infty z^{a-1}e^{-cz}\, dz = \frac{\Gamma(a)}{c^a}$ for $a > 0$, $c > 0$.
$\therefore a = 1/2$, $c = \lambda = \mu + n\theta$

$$\therefore \int_0^\infty z^{1/2-1}e^{-\lambda z}\, dz = \frac{\Gamma(1/2)}{\lambda^{1/2}}.$$

$$\therefore E[Z^{-1/2}] = \lambda \cdot \frac{\Gamma(1/2)}{\lambda^{1/2}} = \lambda^{1-1/2}\Gamma(1/2) = \lambda^{1/2}\Gamma(1/2).$$

Since $\Gamma(1/2) = \sqrt{\pi}$, $\lambda = \mu + n\theta$:

$$E[Z^{-1/2}] = (\mu + n\theta)^{1/2}\sqrt{\pi} = \sqrt{\mu + n\theta} \cdot \sqrt{\pi} = \sqrt{\pi(\mu + n\theta)}.$$

The integral converges because $z^{-1/2}$ is integrable near 0 (since $\int_0^1 z^{-1/2}\, dz < \infty$) and the exponential decay ensures convergence at infinity. Therefore:

$$E\left(\min\{Y, X_1, \ldots, X_n\}^{-1/2}\right) = \sqrt{\pi(\mu + n\theta)}.$$

(ii): $E \max\{Y, X_1, \ldots, X_n\}$

Let $W = \max\{Y, X_1, \ldots, X_n\}$. For a non-negative random variable, the expectation is:

$$E[W] = \int_0^\infty P(W > w)\, dw.$$

$P(W \leq w) = P(Y \leq w, X_1 \leq w, \ldots, X_n \leq w)$, and by independence:

$$P(W \leq w) = P(Y \leq w)\prod_{i=1}^{n} P(X_i \leq w).$$

For $Y \sim \text{Exp}(\mu)$, $P(Y \leq w) = 1 - e^{-\mu w}$, and for each $X_i \sim \text{Exp}(\theta)$, $P(X_i \leq w) = 1 - e^{-\theta w}$. Since the X_i 's are i.i.d.:

$$P(W \leq w) = (1 - e^{-\mu w})(1 - e^{-\theta w})^n.$$

Thus:

$$P(W > w) = 1 - (1 - e^{-\mu w})(1 - e^{-\theta w})^n.$$

Expand $(1 - e^{-\theta w})^n$ using the binomial theorem:

$$(1 - e^{-\theta w})^n = \sum_{k=0}^{n} \binom{n}{k}(-1)^k(e^{-\theta w})^k = \sum_{k=0}^{n} \binom{n}{k}(-1)^k e^{-k\theta w}.$$

So:

$$P(W \le w) = (1 - e^{-\mu w}) \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta w}.$$

Distribute:

$$P(W \le w) = \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta w} - e^{-\mu w} \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta w},$$

$$P(W > w) = 1 - \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta w} + \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-(\mu + k\theta) w}.$$

Separate the $k = 0$ term ($\binom{n}{0} = 1$, $(-1)^0 = 1$):

$$\sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta w} = 1 + \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-k\theta w},$$

$$\sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-(\mu + k\theta) w} = e^{-\mu w} + \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-(\mu + k\theta) w}.$$

Then:

$$P(W > w) = 1 - \left(1 + \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-k\theta w}\right) + e^{-\mu w} + \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-(\mu + k\theta) w},$$

$$= e^{-\mu w} + \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-(\mu + k\theta) w} - \sum_{k=1}^{n} \binom{n}{k} (-1)^k e^{-k\theta w}.$$

Combine the sums:

$$P(W > w) = e^{-\mu w} + \sum_{k=1}^{n} \binom{n}{k} (-1)^k [e^{-(\mu + k\theta) w} - e^{-k\theta w}].$$

Now compute:

$$E[W] = \int_0^\infty e^{-\mu w} \, dw + \sum_{k=1}^{n} \binom{n}{k} (-1)^k \int_0^\infty [e^{-(\mu + k\theta) w} - e^{-k\theta w}] \, dw.$$

- $\int_0^\infty e^{-\mu w} \, dw = \frac{1}{\mu}$,
- $\int_0^\infty e^{-(\mu + k\theta) w} \, dw = \frac{1}{\mu + k\theta}$,
- $\int_0^\infty e^{-k\theta w} \, dw = \frac{1}{k\theta}$.

So:

$$E[W] = E \max\{Y, X_1, \ldots, X_n\} = \frac{1}{\mu} + \sum_{k=1}^{n} \binom{n}{k} (-1)^k \left(\frac{1}{\mu + k\theta} - \frac{1}{k\theta}\right).$$

Q2

- **Service Time for Mr. Eaton**: Let S be the remaining service time for Mr. Eaton, which is exponentially distributed with rate μ. Thus, $S \sim \text{Exp}(\mu)$, and its pdf is $f_S(s) = \mu e^{-\mu s}$ for $s \ge 0$.
- **Patience Times for Waiting Customers**: For each of the n waiting customers, let T_i (where $i = 1, 2, \ldots, n$) be the patience time, which is exponentially distributed with rate θ. Thus, $T_i \sim \text{Exp}(\theta)$, and these are independent of each other and of S.

$\because$ Mr. Eaton is currently being served, and service is sequential, none of the waiting customers' services can begin until Mr. Eaton's service is completed at time S
$\therefore$ Customer i will leave if their patience time T_i is less than S, because their service cannot start before S, and $T_i < S$ means their patience expires before Mr. Eaton's service ends.

Define $M_n = \max\{T_1, T_2, \ldots, T_n\}$. Then, the desired probability is:

$$P(\text{all } n \text{ customers leave}) = P(T_1 < S, T_2 < S, \ldots, T_n < S) = P(M_n < S).$$

For n i.i.d. exponential random variables $T_i \sim \text{Exp}(\theta)$, the CDF of each is:

$$P(T_i \le t) = 1 - e^{-\theta t}, \quad t \ge 0.$$

The CDF of $M_n = \max\{T_1, T_2, \ldots, T_n\}$ is:

$$P(M_n \le t) = P(T_1 \le t, T_2 \le t, \ldots, T_n \le t) = [P(T_1 \le t)]^n = (1 - e^{-\theta t})^n, \quad t \ge 0.$$

Using the law of total probability over S:

$$P(M_n < S) = \int_0^\infty P(M_n < s) f_S(s)\, ds.$$

Substitute the expressions:

- $P(M_n < s) = P(M_n \le s) = (1 - e^{-\theta s})^n$ (since the distribution is continuous),
- $f_S(s) = \mu e^{-\mu s}$.

Thus:

$$P(M_n < S) = \int_0^\infty (1 - e^{-\theta s})^n \mu e^{-\mu s}\, ds.$$

Expand $(1 - e^{-\theta s})^n$ using the binomial theorem:

$$(1 - e^{-\theta s})^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k (e^{-\theta s})^k = \sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta s}.$$

$$\therefore P = \int_0^\infty \left[\sum_{k=0}^{n} \binom{n}{k} (-1)^k e^{-k\theta s} \right] \mu e^{-\mu s}\, ds = \sum_{k=0}^{n} \binom{n}{k} (-1)^k \int_0^\infty \mu e^{-(k\theta + \mu) s}\, ds.$$

The inner integral is:

$$\int_0^\infty \mu e^{-(k\theta+\mu)s}\, ds = \frac{\mu}{k\theta + \mu},$$

since it's the integral of an exponential density with rate $k\theta + \mu$. Thus:

$$P = \sum_{k=0}^{n} \binom{n}{k} (-1)^k \frac{\mu}{k\theta + \mu}.$$

Q3

(a): Service Time is Exactly 10 Minutes

- **A, B, and C begin at** $t = 0$ and, with a fixed service time of 10 minutes, all finish simultaneously at $t = 10$.
- **D starts service at** $t = 10$, when the first clerk becomes available (as A, B, and C all leave at the same time). D's service also takes 10 minutes, so he finishes at $t = 20$.

Track the sequence of events:

- **At** $t = 10$: A, B, and C complete their service and leave the post office. D begins service immediately at one of the freed clerks (it doesn't matter which, since the clerks are identical and the timing is the same).
- **At** $t = 20$: D completes his 10-minute service and leaves.

Since this is a deterministic scenario with no randomness, the outcome is fixed: A and B always leave at $t = 10$, and D leaves at $t = 20$. Thus, it's impossible for A and B to remain in the post office after both C and D have departed. The probability of an impossible event is:

Probability for part (a): 0

(b): Service Times are Exponential with Mean $1/\mu$

Define the service times:

- X_A: Service time for A, $\sim \text{Exp}(\mu)$
- X_B: Service time for B, $\sim \text{Exp}(\mu)$
- X_C: Service time for C, $\sim \text{Exp}(\mu)$
- X_D: Service time for D, $\sim \text{Exp}(\mu)$

All service times are independent and identically distributed (i.i.d.). Denote $T = \min(X_A, X_B, X_C)$: The time when the first of A, B, or C finishes, at which point D begins service.

Since X_A, X_B, X_C are i.i.d. $\text{Exp}(\mu)$, the minimum T follows an exponential distribution with rate 3μ (the rate of the minimum of independent exponentials is the sum of their rates: $\mu + \mu + \mu = 3\mu$).

The problem need A and B to remain in the post office after C and D have left.

Key Scenario: C Leaves First, Then D Leaves Second

The most promising case is when C leaves first, D starts service, and then D leaves before A and B finish.

1. **C leaves first**:
   - Probability that $X_C = \min(X_A, X_B, X_C)$, i.e., $X_C < X_A$ and $X_C < X_B$.
   - Since X_A, X_B, X_C are i.i.d. $\text{Exp}(\mu)$, each has an equal chance of being the smallest.
   - $P(X_C < X_A, X_C < X_B) = 1/3$: among three symmetric exponentials, any one being the minimum has probability $1/3$.
2. **After C leaves at $t = X_C$**:
   - D starts service at $t = X_C$, with service time X_D.
   - D leaves at $T_D = X_C + X_D$.
   - A and B are still in service at $t = X_C$. Due to the memoryless property of exponentials, their *remaining* service times from $t = X_C$ are:
     - $Y_A = X_A - X_C$, $\sim \text{Exp}(\mu)$, since $X_A > X_C$.
     - $Y_B = X_B - X_C$, $\sim \text{Exp}(\mu)$, since $X_B > X_C$.
   - A leaves at $X_C + Y_A$, B at $X_C + Y_B$.
3. **D leaves before A and B**:
   - Now need $T_D < X_A$ and $T_D < X_B$, i.e., $X_C + X_D < X_C + Y_A$ and $X_C + X_D < X_C + Y_B$.
   - This simplifies to $X_D < Y_A$ and $X_D < Y_B$, or equivalently, $X_D < \min(Y_A, Y_B)$.
   - $Y_A, Y_B \sim \text{Exp}(\mu)$ are i.i.d., so $\min(Y_A, Y_B) \sim \text{Exp}(2\mu)$ (rate $\mu + \mu = 2\mu$).
   - $X_D \sim \text{Exp}(\mu)$, independent of Y_A and Y_B.
   - Probability that $X_D < \min(Y_A, Y_B)$:
     - For $U \sim \text{Exp}(\lambda_1)$, $V \sim \text{Exp}(\lambda_2)$, independent, $P(U < V) = \lambda_1/(\lambda_1 + \lambda_2)$.
     - Here, X_D rate = μ, $\min(Y_A, Y_B)$ rate = 2μ, so:

$$P(X_D < \min(Y_A, Y_B)) = \frac{\mu}{\mu + 2\mu} = \frac{\mu}{3\mu} = \frac{1}{3}$$

   - Alternatively, among Y_A, Y_B, X_D (three i.i.d. $\text{Exp}(\mu)$), the probability any one is the smallest is $1/3$, confirming $P(X_D < \min(Y_A, Y_B)) = 1/3$.
4. **Total probability**:
   - Events are independent due to memorylessness: $P(\text{C first}) \times P(\text{D next before A and B})$.
   - $P = \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$.

Only when C leaves first ($T = X_C$), then D leaves second ($X_C + X_D < X_A, X_B$), are A and B still in service when both C and D have left (at $t = X_C + X_D$). No other order satisfies this.Thus:

Probability for part (b): 1/9

Q4

(a): Finding $E[X]$

Denote T as the random variable of the time between the departure of the last train and the arrival of the next train, which is uniformly distributed over $(0, 1)$. Since $T \sim \text{Uniform}(0, 1)$, its pdf is $f_T(t) = 1$ for $0 < t < 1$, and its expected value is:

$$E[T] = \int_0^1 t \cdot 1 \, dt = \left[\frac{t^2}{2}\right]_0^1 = \frac{1}{2} - 0 = \frac{1}{2}.$$

Passengers arrive according to a Poisson process with rate $\lambda = 7$ per hour. The time interval T is random, so the number of passengers X who arrive between trains, given $T = t$, is:

$$X \mid T = t \sim \text{Poisson}(7t),$$

because the rate is 7 per hour, and over time t hours, the expected number of arrivals is $7t$.

According to the **law of total expectation**:

$$E[X] = E[E[X \mid T]].$$

For a Poisson random variable with parameter μ, the expected value is μ. Thus, if $X \mid T = t \sim \text{Poisson}(7t)$, then:

$$E[X \mid T = t] = 7t.$$

Since this is a function of the random variable T:

$$E[X] = E[E[X \mid T]] = E[7T] = 7E[T] = 7 \cdot \frac{1}{2} = \frac{7}{2}.$$

(b): Finding $\text{Var}(X)$

According to the **law of total variance**:

$$\text{Var}(X) = E[\text{Var}(X \mid T)] + \text{Var}(E[X \mid T]).$$

Since $X \mid T = t \sim \text{Poisson}(7t)$:

$$\text{Var}(X \mid T = t) = 7t.$$

Thus, $\text{Var}(X \mid T) = 7T$. What's more, the expectation:

$$E[\text{Var}(X \mid T)] = E[7T] = 7E[T] = 7 \cdot \frac{1}{2} = \frac{7}{2}.$$

From part (a):

$$E[X \mid T] = 7T.$$

So:

$$\text{Var}(E[X \mid T]) = \text{Var}(7T) = 7^2 \,\text{Var}(T) = 49 \,\text{Var}(T).$$

For $T \sim \text{Uniform}(0, 1)$, the variance is:

$$\text{Var}(T) = \frac{(b-a)^2}{12} = \frac{(1-0)^2}{12} = \frac{1}{12}.$$

Thus:

$$\text{Var}(7T) = 49 \cdot \frac{1}{12} = \frac{49}{12}.$$

Sum the two components:

$$\text{Var}(X) = E[\text{Var}(X \mid T)] + \text{Var}(E[X \mid T]) = \frac{7}{2} + \frac{49}{12} = \frac{91}{12}.$$

Q5

(a): Probability of at least one of types 1, 2, 3 and none of type 4 by time t

Since the overall process is a Poisson process with rate λ, and each car is independently type i with probability p_i: split this into four independent Poisson processes—one for each car type. The rate for type i is the overall rate λ multiplied by the probability p_i, so:

- Number of type i cars by time t, denoted $N_i(t)$, follows a Poisson distribution with parameter $\lambda p_i t$.

These processes are independent because the car arrivals follow a Poisson process, and the type assignments are independent.

Define the events:

- $N_1(t) \geq 1$: at least one type 1 car.
- $N_2(t) \geq 1$: at least one type 2 car.
- $N_3(t) \geq 1$: at least one type 3 car.
- $N_4(t) = 0$: exactly zero type 4 cars.

Since the processes are independent:

$$P(N_1(t) \geq 1, N_2(t) \geq 1, N_3(t) \geq 1, N_4(t) = 0) = P(N_1(t) \geq 1) \cdot P(N_2(t) \geq 1) \cdot P(N_3(t) \geq 1) \cdot P(N_4(t) = 0)$$

For a Poisson random variable X with parameter μ:

- $P(X = 0) = e^{-\mu}$,
- $P(X \geq 1) = 1 - P(X = 0) = 1 - e^{-\mu}$.

Apply this:

- $N_1(t) \sim \text{Poisson}(\lambda p_1 t)$, so $P(N_1(t) \geq 1) = 1 - e^{-\lambda p_1 t}$,
- $N_2(t) \sim \text{Poisson}(\lambda p_2 t)$, so $P(N_2(t) \geq 1) = 1 - e^{-\lambda p_2 t}$,
- $N_3(t) \sim \text{Poisson}(\lambda p_3 t)$, so $P(N_3(t) \geq 1) = 1 - e^{-\lambda p_3 t}$,
- $N_4(t) \sim \text{Poisson}(\lambda p_4 t)$, so $P(N_4(t) = 0) = e^{-\lambda p_4 t}$.

Thus, the probability is:

$$(1 - e^{-\lambda p_1 t})(1 - e^{-\lambda p_2 t})(1 - e^{-\lambda p_3 t})e^{-\lambda p_4 t}$$

(b): Probability that 4 out of 6 cars of type 1 or 2 are type 1

Define:

- $N_1(t)$: number of type 1 cars by time t, $\sim \text{Poisson}(\lambda p_1 t)$,
- $N_2(t)$: number of type 2 cars by time t, $\sim \text{Poisson}(\lambda p_2 t)$.

The total number of type 1 or type 2 cars is $N_1(t) + N_2(t)$, and since the sum of independent Poisson random variables is Poisson:

$$N_1(t) + N_2(t) \sim \text{Poisson}(\lambda p_1 t + \lambda p_2 t) = \text{Poisson}(\lambda(p_1 + p_2)t)$$

It's now conditioned on $N_1(t) + N_2(t) = 6$, and we want:

$$P(N_1(t) = 4 \mid N_1(t) + N_2(t) = 6)$$

Since $N_1(t) = 4$ and the total is 6, it follows that $N_2(t) = 6 - 4 = 2$.

According to the Poisson thinning theorem:

- $N_1(t) \sim \text{Poisson}(\lambda p_1 t)$
- $N_2(t) \sim \text{Poisson}(\lambda p_2 t)$
- Total parameter: $\lambda(p_1 + p_2)t$,
- Success probability: $p = \frac{\lambda p_1 t}{\lambda(p_1+p_2)t} = \frac{p_1}{p_1+p_2}$.

Given $N_1(t) + N_2(t) = 6$, $N_1(t) \sim \text{Binomial}(6, \frac{p_1}{p_1+p_2})$. The probability of exactly 4 successes is:

$$P(N_1(t) = 4) = \binom{6}{4}\left(\frac{p_1}{p_1+p_2}\right)^4\left(1-\frac{p_1}{p_1+p_2}\right)^2 = \binom{6}{4}\left(\frac{p_1}{p_1+p_2}\right)^4\left(\frac{p_2}{p_1+p_2}\right)^2$$